

AIOps Assignment2

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Question 1

3) The naive build is 627 MB, while the multi-stage build is just 177 MB, a reduction of 71.8 %. The multi-stage image is smaller because its builder stage (the full python:3.11 base and pip's download cache from installing pandas, scikit-learn, fastapi) is entirely discarded after use. Only the finished venv folder of installed packages, main.py, and model.joblib get copied into the final stage.

Question 2

4) Docker compose allows for configuring multiple containers with limited commands. Without compose, you'd have to manually set up networking between the two containers, tart them in the right order, and re-type both docker run commands every time. Docker compose allows to handle multiple containers on a single machine, whereas kubernetes is used to run across many machines, be it for scalability or reliability.

Question 3

1) With 4 CPUs available (2 nodes x 2 CPUs), parallelism of 4 uses the full budget per wave, completing all 8 in 2 waves of 4 concurrent pods

4) I'd raise parallelism from 4 to 6. Since each pod reserves exactly 1 CPU (requests equal to limits), having 6 CPUs available instead of 4 means 6 pods can now run at the same time rather than 4, so raising the parallelism value lets us use the extra CPU capacity instead of leaving it idle.

Question 4

2) The ReplicaSet controller is responsible. It continuously reconciles the observed pod count matching its label selector against spec.replicas (2 here). Deletion drops the count to 1, and the next reconcile creates a replacement.

3) A Job runs its pods until they finish and then simply stops, it doesn't run forever. If our API had been deployed as a Job, updating the pod template to point at the new image tag and reapplying it wouldn't have triggered any rollout at all. There would have been no pod by pod handover by the readiness probe. Reaching the new version would have meant deleting the whole Job and creating a new one.

4) A Deployment is built for work that has no notion of completion, and must stay available continuously, which is the spam-detection API. It keeps a fixed number of replicas running forever, replacing any that die on their own (self-healing) and swapping new versions gradually so the available pod count never drops to zero (rolling updates).

A Job is built for work that has a notion of completion, like validating 8 CSV shards. Its only concern is confirming each pod's task completed successfully once